

NANAKWA SHOAL BUOY 46181, Canada

WMO: 996520

Lat: 53.830N

Lon: 128.830W

Elev: 10

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.7	(c) 18.8	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 36.2	(k) 16.8	(l) 34.8	(m) 17.3	(n) 27.6	(o) 40	(p) 0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.7	(c) 73.1	(d) N/A	(e) 70.0	(f) N/A	(g) 67.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.3	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 29.9	(b) 26.7	(c) 23.5	(e) 11.5	(f) 80.3	(g) 5.5	(h) 4.1	(i) 7.6	(j) 83.2	(k) 4.3	(l) 85.6	(m) 1.2	(n) 88.0	(o) -2.8	(p) 91.0		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	33.8	36.0	38.5	44.1	51.3	56.8	60.5	60.8	55.0	47.0	38.4	34.2
	DBStd	11.04	7.54	4.96	4.79	4.30	5.03	4.55	4.71	3.77	3.79	4.13	5.39	5.95
	HDD50	2430	502	391	357	188	43	1	0	0	5	108	347	488
	HDD65	6810	967	811	822	628	424	251	158	140	302	557	797	953
	CDD50	1129	0	0	0	9	84	205	326	336	153	16	0	0
	CDD65	35	0	0	0	0	0	5	19	11	0	0	0	0
	CDH74	71	0	0	0	0	0	13	44	14	0	0	0	0
	CDH80	7	0	0	0	0	0	1	5	1	0	0	0	0
Wind	WSAvg	10.2	13.2	10.9	9.9	8.5	9.0	9.8	9.9	9.1	8.1	9.4	11.6	13.4
Precipitation	PrecAvg	98.9	12.8	8.0	7.3	5.6	3.7	3.1	3.0	4.6	8.1	14.9	14.3	15.1
	PrecMax	119.7	18.9	13.7	12.0	11.8	7.3	5.2	5.4	8.0	24.2	24.2	22.3	28.7
	PrecMin	82.3	5.4	2.6	3.0	3.0	1.7	1.5	0.8	1.7	2.5	4.3	7.0	9.7
	PrecStd	11.4	3.4	3.4	2.4	2.4	1.9	1.1	1.2	2.1	5.3	4.3	4.0	5.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	46.7	49.7	62.1	68.8	75.9	78.7	75.7	68.5	57.7	50.7	47.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	44.4	44.3	47.4	56.9	64.9	70.4	74.1	71.4	64.6	55.6	48.3	44.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	42.6	42.8	45.5	53.1	61.7	66.6	70.6	68.8	62.0	53.9	46.5	42.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	40.9	41.4	43.9	50.1	58.7	63.6	67.4	66.5	59.9	52.4	44.6	40.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.9	7.0	8.3	9.2	8.2	7.9	7.7	6.4	5.5	5.1	4.8
		MCDBR	7.3	7.3	9.1	13.5	14.2	14.5	14.0	12.1	10.4	7.4	7.0	7.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.303	0.309	0.348	0.387	0.420	0.415	0.384	0.383	0.359	0.347	0.321	0.293	
	taud	2.213	2.288	2.179	2.121	2.148	2.265	2.409	2.430	2.485	2.407	2.253	2.192	
	Ebn at Noon	203	248	260	265	262	265	272	266	259	232	194	180	
	Edh at Noon	20	26	36	43	45	40	34	32	27	23	19	17	
All-Sky Solar Radiation	RadAvg	164	410	656	1086	1424	1513	1457	1293	820	451	200	135	
	RadStd	25	87	121	87	179	145	212	109	116	88	41	24	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (15 neighbors)		N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page